

## NEGATIVITY LEMMA

*Setting and conventions.* Fix an algebraically closed field  $k$  of arbitrary characteristic. All varieties are integral separated schemes of finite type over  $k$ , and all morphisms are  $k$ -morphisms. On a normal variety, an  $\mathbb{R}$ -Cartier divisor is a finite real linear combination of Cartier divisors, viewed also as a Weil  $\mathbb{R}$ -divisor. Effectivity and support refer to its prime-divisor coefficients. For a proper morphism,  $f_*D$  denotes the pushforward of this Weil divisor: a prime divisor is sent to its image with the function-field degree when the image is a prime divisor, and to zero otherwise.

For a Cartier divisor  $A$  and a complete integral curve  $C \subset X$ , write  $A \cdot C = \deg \nu^*(\mathcal{O}_X(A)|_C)$ , where  $\nu: \tilde{C} \rightarrow C$  is the normalization; extend this intersection number  $\mathbb{R}$ -linearly. A divisor  $D$  is  $f$ -nef if  $D \cdot C \geq 0$  for every such curve contracted by  $f$ . A Cartier divisor  $A$  is  $f$ -ample if  $\mathcal{O}_X(A)$  is relatively ample. In particular, a projective embedding  $i: X \hookrightarrow \mathbf{P}_Y^n$  supplies the relatively very ample line bundle  $i^*\mathcal{O}_{\mathbf{P}_Y^n}(1)$ . Fibers and containments in supports below are understood set-theoretically. No separability or characteristic-zero hypothesis is imposed.

### 1. DETAILED ARGUMENT

**Lemma 1.1.** *Let  $f: X \rightarrow Y$  be a projective birational morphism between normal varieties, with  $Y$  affine. Then there exists an effective Cartier divisor  $E$  on  $X$  such that  $-E$  is  $f$ -ample.*

*Proof.* Since  $f$  is projective, we can choose an  $f$ -ample Cartier divisor  $D$  on  $X$ . Put  $\mathcal{F} = f_*\mathcal{O}_X(-D)$ . Since  $f$  is projective,  $\mathcal{F}$  is coherent. Since  $f$  is birational, it is an isomorphism over a dense open subset of  $Y$ , on which  $\mathcal{F}$  is invertible. Thus  $\mathcal{F}$  has generic rank one; in particular,  $\mathcal{F} \neq 0$ .

Since  $Y$  is affine, a nonzero quasi-coherent sheaf on  $Y$  has a nonzero global section. Hence we may choose

$$0 \neq s \in H^0(Y, \mathcal{F}) = H^0(X, \mathcal{O}_X(-D)).$$

Viewing  $\mathcal{O}_X(-D)$  as a subsheaf of the sheaf of rational functions, we regard  $s$  as a nonzero rational function on  $X$ . Set

$$E = -D + \operatorname{div}(s).$$

The condition  $s \in H^0(X, \mathcal{O}_X(-D))$  says precisely that  $E$  is an effective Cartier divisor. Indeed, if  $d$  is a local equation of  $D$ , then  $s/d$  is a nonzero regular function and is a local equation of  $E$ . Finally,  $-E \sim D$ , so  $-E$  is  $f$ -ample.  $\square$

**Lemma 1.2.** *Let  $f: X \rightarrow Y$  be a projective birational morphism between varieties, with  $Y$  normal, and let  $E$  be an effective Cartier divisor on  $X$  such that  $-E$  is  $f$ -ample. Then  $\text{Supp } E$  contains the exceptional locus  $\text{Exc}(f)$ .*

*Proof.* Since  $Y$  is normal,  $\text{Exc}(f)$  is covered by complete integral curves contracted by  $f$ . For every such curve  $C$ , the  $f$ -ampleness of  $-E$  gives  $E \cdot C < 0$ . Since  $E$  is effective, this forces  $C \subseteq \text{Supp } E$ . Hence  $\text{Exc}(f) \subseteq \text{Supp } E$ .  $\square$

**Theorem 1.3** (Projective negativity lemma). *Let  $f: X \rightarrow Y$  be a projective birational morphism between normal varieties. Let  $D$  be an  $\mathbb{R}$ -Cartier divisor on  $X$  such that  $-D$  is nef over  $Y$ . Then:*

- (1)  $D$  is effective if and only if  $f_*D$  is effective.
- (2) If  $D$  is effective, then for every  $y \in Y$ , either

$$f^{-1}(y) \cap \text{Supp } D = \emptyset \quad \text{or} \quad f^{-1}(y) \subset \text{Supp } D.$$

*Proof of Theorem 1.3.* Both assertions are local on  $Y$ , so we may assume that  $Y$  is affine.

(1) One direction is trivial. For the other direction, assume  $f_*D \geq 0$ . Since  $f$  is birational, every prime divisor with negative coefficient in  $D$  is  $f$ -exceptional.

Suppose, for a contradiction, that  $D$  is not effective. Choose an effective Cartier divisor  $E$  on  $X$  such that  $-E$  is  $f$ -ample, as in Lemma 1.1. By Lemma 1.2, the coefficient  $\text{coeff}_P(E) > 0$  whenever  $\text{coeff}_P(D) < 0$ . Set

$$e := \max_{\text{coeff}_P(D) < 0} \frac{-\text{coeff}_P(D)}{\text{coeff}_P(E)} > 0.$$

Then  $D + eE \geq 0$ , and we can find an  $f$ -exceptional prime divisor  $F$  such that  $F \not\subset \text{Supp}(D + eE)$ . Choose a closed point  $x \in F \setminus \text{Supp}(D + eE)$  and a curve  $C \subset F$  through  $x$  contracted by  $f$ . Then  $C \not\subset \text{Supp}(D + eE)$ , so  $(D + eE) \cdot C \geq 0$ . On the other hand,

$$(D + eE) \cdot C = D \cdot C + e(E \cdot C) < 0,$$

because  $-D$  is  $f$ -nef,  $-E$  is  $f$ -ample, and  $e > 0$ . This is a contradiction, so  $D \geq 0$ .

(2) Assume  $D \geq 0$ . Since  $f$  is projective and birational and  $Y$  is normal, its fibers are connected by Stein factorization. Suppose  $f^{-1}(y) \cap \text{Supp } D \neq \emptyset$ . If  $f^{-1}(y) \not\subset \text{Supp } D$ , connectedness gives a curve  $C \subset f^{-1}(y)$  meeting  $\text{Supp } D$  but not contained in it. Then  $D \cdot C > 0$ , contradicting the  $f$ -nefness of  $-D$ . Thus  $f^{-1}(y) \subset \text{Supp } D$ .  $\square$

**Theorem 1.4** (Proper negativity lemma). *Let  $f: X \rightarrow Y$  be a proper birational morphism between normal varieties. Let  $D$  be an  $\mathbb{R}$ -Cartier divisor on  $X$  such that  $-D$  is nef over  $Y$ . Then:*

- (1)  $D$  is effective if and only if  $f_*D$  is effective.

(2) If  $D$  is effective, then for every  $y \in Y$ , either

$$f^{-1}(y) \cap \text{Supp } D = \emptyset \quad \text{or} \quad f^{-1}(y) \subset \text{Supp } D.$$

*Proof.* By Chow's lemma and normalization, choose a proper birational morphism  $\pi: X' \rightarrow X$  with  $X'$  normal such that  $f' = f \circ \pi: X' \rightarrow Y$  is projective. Set  $D' = \pi^* D$ . By the projection formula,  $-D'$  is  $f'$ -nef, and

$$\pi_* D' = D, \quad f'_* D' = f_* D.$$

If  $f_* D \geq 0$ , Theorem 1.3(1) gives  $D' \geq 0$ , hence  $D = \pi_* D' \geq 0$ . The converse is immediate.

If  $D \geq 0$ , then  $D' \geq 0$  and  $\text{Supp } D' = \pi^{-1}(\text{Supp } D)$ . By Theorem 1.3(2), each fiber of  $f'$  is either disjoint from  $\text{Supp } D'$  or contained in it. Since  $f'^{-1}(y) \rightarrow f^{-1}(y)$  is surjective, the same holds for  $f$  and  $D$ .  $\square$

## 2. A CONCISE PROOF

*Proof of Theorem 1.4.* We may work over an affine open subset of  $Y$ . By Chow's lemma and normalization, choose  $\pi: X' \rightarrow X$  proper birational with  $X'$  normal and  $f \circ \pi$  projective. The projection formula preserves relative nefness, and

$$\pi_* \pi^* D = D, \quad \text{Supp}(\pi^* D) = \pi^{-1}(\text{Supp } D) \quad \text{if } D \geq 0.$$

Thus both assertions descend from  $X'$  to  $X$ , and we may assume that  $f$  is projective.

(1) Suppose  $f_* D \geq 0$  but  $D \not\geq 0$ . Choose an  $f$ -ample Cartier divisor  $A$ . The coherent sheaf  $f_* \mathcal{O}_X(-A)$  has generic rank one and hence, since  $Y$  is affine, a nonzero global section  $s$ . Then  $E = -A + \text{div}(s) \geq 0$  and  $-E$  is  $f$ -ample. Every contracted curve  $C$  satisfies  $E \cdot C < 0$ , so  $C \subset \text{Supp } E$ . Such curves cover  $\text{Exc}(f)$ ; hence  $\text{Exc}(f) \subset \text{Supp } E$ .

All negative components of  $D$  are  $f$ -exceptional. Set

$$e := \max_{\text{coeff}_P(D) < 0} \frac{-\text{coeff}_P(D)}{\text{coeff}_P(E)} > 0.$$

Then  $D + eE \geq 0$ , and some  $f$ -exceptional prime divisor  $F$  is not contained in  $\text{Supp}(D + eE)$ . Choose a contracted curve  $C \subset F$  not contained in this support. We obtain

$$0 \leq (D + eE) \cdot C = D \cdot C + e(E \cdot C) < 0,$$

a contradiction. The reverse implication is immediate.

(2) Assume  $D \geq 0$  and  $f^{-1}(y) \cap \text{Supp } D \neq \emptyset$ . The fiber is connected by Stein factorization. If it is not contained in  $\text{Supp } D$ , there is a curve in the fiber meeting  $\text{Supp } D$  but not contained in it. Its intersection with  $D$  is positive, contrary to the  $f$ -nefness of  $-D$ . Thus  $f^{-1}(y) \subset \text{Supp } D$ .  $\square$

## 3. EXTENSION TO QUASI-PROJECTIVE BASES

Lemma 1.1 remains valid when the affine hypothesis on  $Y$  is replaced by quasi-projectivity: for a projective birational morphism  $f: X \rightarrow Y$  between normal varieties with  $Y$  quasi-projective, there is an effective Cartier divisor  $E$  such that  $-E$  is  $f$ -ample. This strengthening also requires no Riemann–Roch.

*Proof.* Choose an  $f$ -ample line bundle  $\mathcal{L}$  and put  $\mathcal{F} = f_*\mathcal{L}^{-1}$ . As in Lemma 1.1,  $\mathcal{F}$  is coherent of generic rank one, so  $\mathcal{F} \neq 0$ .

Choose a very ample Cartier divisor  $H$  on  $Y$ . By the global-generation criterion for ampleness,  $\mathcal{F} \otimes \mathcal{O}_Y(mH)$  is generated by global sections for  $m \gg 0$ . It is nonzero, so it has a nonzero global section. The projection formula gives

$$\begin{aligned} 0 \neq s &\in H^0(Y, \mathcal{F} \otimes \mathcal{O}_Y(mH)) \\ &= H^0(X, \mathcal{L}^{-1} \otimes \mathcal{O}_X(mf^*H)). \end{aligned}$$

Its zero divisor  $E$  satisfies

$$\mathcal{O}_X(-E) \simeq \mathcal{L} \otimes f^*\mathcal{O}_Y(-mH).$$

Tensoring with a line bundle pulled back from the base preserves relative ampleness, so  $-E$  is again  $f$ -ample. If  $Y$  is projective, then  $X$  is projective. Taking  $\mathcal{L} = \mathcal{O}_X(A)$  for any very ample divisor  $A$  on  $X$  gives  $E \sim mf^*H - A$  and hence the decomposition

$$mf^*H \sim A + E, \quad -E \sim_f A.$$

Lemma 1.2 further shows that every exceptional prime divisor lies in  $\text{Supp } E$ . The only additional input is global generation after a sufficiently large ample twist. No Riemann–Roch or section-growth estimate is needed.  $\square$

**Remark 3.1.** The affine hypothesis is essential to the *untwisted argument*, not to the conclusion over a quasi-projective base. A nonzero coherent sheaf on a projective variety need not have any global sections. For example, with  $f = \text{id}_{\mathbf{P}^1}$  and  $\mathcal{L} = \mathcal{O}_{\mathbf{P}^1}(1)$ , one has  $f_*\mathcal{L}^{-1} = \mathcal{O}_{\mathbf{P}^1}(-1) \neq 0$  but  $H^0(X, \mathcal{L}^{-1}) = 0$ . The twist by  $mH$  repairs precisely this failure. On an arbitrary normal base, a global very ample  $H$  need not exist; for the negativity lemma it suffices to apply the affine construction on each affine open subset of the base.

**Remark 3.2.** For comparison, the projective decomposition also has the following proof, making explicit the section-growth argument usually used in Kodaira’s lemma. Assume that  $Y$  is projective, put  $n = \dim X = \dim Y \geq 1$ , choose very ample Cartier divisors

$H$  on  $Y$  and  $A$  on  $X$ , and set  $M = f^*H$ . Since  $Y$  is normal and  $f$  is proper birational,  $f_*\mathcal{O}_X = \mathcal{O}_Y$ . Thus

$$h^0(X, \mathcal{O}_X(mM)) = h^0(Y, \mathcal{O}_Y(mH)) = \frac{H^n}{n!}m^n + O(m^{n-1}), \quad H^n > 0.$$

The last equality follows from Serre vanishing and the Hilbert polynomial of the very ample divisor  $H$ . In particular,  $M$  is big; it need not be ample.

Choose an effective member  $T \in |A|$ . It is a Cartier divisor on the integral variety  $X$ , and  $\dim T = n - 1$ . The coherent sheaf  $\mathcal{G} = f_*\mathcal{O}_T$  is supported on  $f(T)$ , whose dimension is at most  $n - 1$ . By the projection formula and the Hilbert polynomial of  $\mathcal{G}$  on  $Y$ ,

$$\begin{aligned} h^0(T, \mathcal{O}_T(mM)) &= h^0(Y, \mathcal{G} \otimes \mathcal{O}_Y(mH)) \\ &= O(m^{n-1}). \end{aligned}$$

No normality or irreducibility of  $T$  is needed. Taking global sections of the exact sequence

$$0 \longrightarrow \mathcal{O}_X(mM - T) \longrightarrow \mathcal{O}_X(mM) \longrightarrow \mathcal{O}_T(mM) \longrightarrow 0$$

gives

$$\begin{aligned} h^0(X, \mathcal{O}_X(mM - A)) &\geq h^0(X, \mathcal{O}_X(mM)) - h^0(T, \mathcal{O}_T(mM)) \\ &\geq \frac{H^n}{n!}m^n - O(m^{n-1}) > 0 \end{aligned}$$

for  $m \gg 0$ . We only use left exactness here; surjectivity of the restriction map is unnecessary. A nonzero section therefore gives an effective Cartier divisor  $E \sim mM - A$ . Equivalently,

$$f^*H \sim_{\mathbb{Q}} \frac{1}{m}A + \frac{1}{m}E, \quad -E \sim_f A.$$

This is the requested decomposition into an ample divisor and an effective divisor. Lemma 1.2 shows that  $\text{Supp } E$  contains every exceptional prime divisor. When  $n = 0$ , the assertion is immediate with  $E = 0$ .

The essential numerical input in this second proof is that the space of sections on  $X$  grows like  $m^n$ , whereas the space of restrictions to  $T$  grows at most like  $m^{n-1}$ . It is often expressed as an asymptotic Riemann–Roch argument. However, full Riemann–Roch is *not logically necessary*: the Hilbert polynomial and Serre vanishing on  $Y$  already give both estimates, and the first proof needs only global generation after an ample twist. Riemann–Roch alone computes an Euler characteristic, not  $h^0$ ; a proof invoking it must also justify the relevant cohomological estimates. In particular, one cannot apply Serre vanishing on  $X$  merely because  $f^*H$  is big and nef. Neither construction uses the Hodge index theorem, and  $E$  may have nonexceptional components.